

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _K.YAMINI(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Yamini(192572181)

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

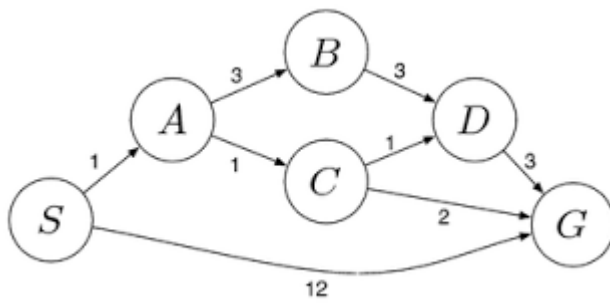
- I. What are the percept's for this agent?
- II. Characterize the operating environment.
- III. What are the actions the agent can take?
- IV. How can one evaluate the performance of the agent?
- V. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem

1. Problem Understanding

We are given:

- A **4-gallon jug** and a **3-gallon jug**
- No measurement markings on either jug
- Unlimited water supply (pump)

Goal:

To measure **exactly 2 gallons of water in the 4-gallon jug**

Allowed Operations:

- Fill a jug completely
- Empty a jug
- Pour water from one jug to another

2. Method Selection

This problem is solved using a **State Space Search approach**.

Each state represents the amount of water in both jugs.

We systematically apply operations to reach the goal state.

We use:

- State representation
- Step-by-step transitions (similar to BFS)

3. Solution Process

State Representation

Let a state be (x, y)

Where:

- x = water in 4L jug
- y = water in 3L jug

Initial State = $(0,0)$

Goal State = $(2,0)$

Step-by-Step Execution

Step	Action Performed	State (4L, 3L)	Explanation
1	Fill 3L jug	(0,3)	Start by filling smaller jug
2	Pour 3L $\rightarrow$ 4L	(3,0)	Transfer all water
3	Fill 3L jug again	(3,3)	Refill
4	Pour 3L $\rightarrow$ 4L	(4,2)	Only 1L fits in 4L jug
5	Empty 4L jug	(0,2)	Remove water from 4L
6	Pour 2L $\rightarrow$ 4L	(2,0)	Transfer remaining water

4. Accuracy of Calculation

- All steps follow jug capacity limits

- No overflow occurs
- Exact water transfer is maintained

Final state is verified: **(2,0)**

4L jug contains exactly **2 gallons**

5. Interpretation of Result

This solution demonstrates how **Artificial Intelligence uses search techniques** to solve problems where direct measurement is not possible.

It shows:

- Logical reasoning
- Systematic state transitions
- Goal-oriented problem solving

Final Answer

Exactly 2 gallons of water is obtained in the 4-gallon jug.

2. Mars Rover as Intelligent Agent

1. Problem Understanding

The **Mars Rover** is an **autonomous intelligent agent** used by space agencies to explore the surface of Mars.

It performs tasks like:

- Navigating terrain
- Collecting soil and rock samples
- Analyzing environmental conditions

It operates in a **remote, uncertain, and dynamic environment**, without continuous human control.

2. Method Selection

To analyze this agent, we use the **PEAS Model** and **Agent Architecture classification**.

PEAS = Performance measure, Environment, Actuators, Sensors

Helps to clearly define how the agent works

3. Solution Process

i. Percepts (Sensors / Inputs)

The rover perceives the environment using:

- Camera images (terrain, rocks, obstacles)
- Temperature sensors
- Atmospheric pressure sensors
- Chemical sensors (soil composition)
- Distance/obstacle sensors

ii. Characterization of Environment

The Mars Rover operates in:

- **Partially Observable** → cannot sense entire environment

- **Stochastic** → uncertain outcomes (dust storms, terrain)
- **Sequential** → current action affects future actions
- **Dynamic** → environment may change over time
- **Continuous** → infinite possible states

iii. Actions (Actuators)

The rover can perform:

- Move forward/backward
- Turn left/right
- Drill the surface
- Collect soil/rock samples
- Analyze samples
- Send data to Earth

iv. Performance Measure

The effectiveness of the rover is evaluated based on:

- Accuracy of collected scientific data
- Number of samples collected
- Distance explored
- Energy efficiency (battery usage)
- Safety (avoiding obstacles/damage)
- Mission success rate

v. Suitable Agent Architecture

Utility-Based Agent

Justification

- The rover must choose actions that maximize overall benefit

- It considers:
 - Scientific value of data
 - Risk level
 - Energy consumption

Thus, it selects the **best possible action based on utility (performance)**

Working Model (Diagram Explanation)

Sensors → Perception → Decision Making → Actions

4. Accuracy

- All components correctly mapped to AI agent structure
- Proper classification of environment and agent type

5. Interpretation of Results

The Mars Rover demonstrates how **AI agents operate autonomously in real-world environments**, making intelligent decisions under uncertainty.

It is a real-world example of:

- Intelligent agents
- Autonomous systems
- Decision-making under constraints

Final Answer

The Mars Rover is best modeled as a **Utility-Based Intelligent Agent**, capable of perceiving its environment, making optimal decisions, and performing actions to achieve mission goals efficiently.

3. 8-Queens Problem

1. Problem Understanding

The objective is to place **8 queens on an 8×8 chessboard** such that:

- No two queens attack each other
- No two queens share:
 - Same row
 - Same column
 - Same diagonal

2. Method Selection

This is a **Constraint Satisfaction Problem (CSP)**.

Best method: **Backtracking Search (DFS)**

Because:

- It systematically explores possible configurations
- Eliminates invalid states early

3. Solution Process

Problem Formulation

State Space

- All possible arrangements of queens on board

Initial State

- Empty board (no queens placed)

Goal State

- 8 queens placed without any conflict

Operators (Actions)

- Place one queen in each column

Constraints

- No same row
- No same column
- No same diagonal

Algorithm Steps (Backtracking)

1. Start from column 1
2. Place queen in a safe row
3. Move to next column
4. Check for conflicts:
 - Row conflict
 - Diagonal conflict
5. If conflict occurs → **Backtrack**
6. Try next possible row
7. Repeat until all 8 queens placed

4. Accuracy

- Constraints strictly followed
- No attacking queens
- Valid solution ensured

5. Interpretation

This problem demonstrates:

- **Search strategies in AI**
- **Constraint satisfaction techniques**
- Efficient pruning using backtracking

Final Answer

The 8-Queens problem is solved using **Backtracking Search**, ensuring all constraints are satisfied to reach a valid configuration.

4. OLA Cab Problem

1. Problem Understanding

A customer uses OLA app to travel from source to destination by selecting a cab type such as:

- Mini
- Micro
- Sedan
- Shared
- Prime

The system must help select the best cab.

2. Method Selection

Goal-Based Agent

Reason

- Clear goal: reach destination
- Agent evaluates options to achieve goal

3. Solution Process

Agent Working

Inputs (Percepts)

- Source location
- Destination
- Available cabs

Decision Factors

- Cost
- Travel time
- Comfort

Steps

1. User enters source & destination
2. System retrieves available cab options
3. Calculates fare, time, comfort
4. Compares options
5. Selects best cab
6. Books ride

Pseudocode

FUNCTION OLA_AGENT(start, destination):

 DEFINE goal = reach destination

 GET cab_options

 FOR each cab:

 CALCULATE cost, time, comfort

 SELECT optimal cab

 BOOK cab

 RETURN success

4. Accuracy

- Decision is based on valid parameters
- Goal satisfaction ensured

5. Interpretation

This shows how **AI agents are used in real-world applications** like ride booking systems to make intelligent decisions.

Final Answer

The OLA cab system uses a **Goal-Based Agent** to select the best cab and achieve the user's travel objective efficiently.

5. Uniform Cost Search

1. Problem Understanding

A logistics company wants to find the **minimum cost path** from:

- Start node (S)
- Goal node (G)

Each path has a cost (fuel + time).

2. Method Selection

Uniform Cost Search (UCS)

Reason

- Expands lowest-cost node first
- Guarantees optimal solution

3. Solution Process

Algorithm Steps

1. Start at node S with cost = 0
2. Insert into **priority queue**
3. Remove node with lowest cost
4. If goal reached → stop
5. Else expand neighbors
6. Add neighbors with updated cost
7. Repeat

Pseudocode

FUNCTION UCS(start, goal):

 CREATE priority_queue

 ADD (start, cost=0)

WHILE queue not empty:

node = REMOVE lowest cost node

IF node == goal:

RETURN path

FOR each neighbor:

new_cost = current_cost + edge_cost

ADD neighbor to queue

RETURN failure

Example Explanation

If:

- Path 1 → cost = 10
- Path 2 → cost = 7

UCS selects **Path 2 (minimum cost)**

4. Accuracy

- Cost calculated correctly
- Always selects lowest-cost node
- Ensures optimal path

5. Interpretation

Used in:

- Logistics
- GPS navigation
- Network routing

Ensures **efficient and cost-effective solutions**

Final Answer

Uniform Cost Search finds the **least-cost optimal path** by expanding nodes in order of increasing cost.